| Question | | | Answer | Marks | AOs | | Guidance |
|---|---|---|---|---|---|---|---|
| 6 | i | | $f(-1)=4\times(-1)^3-3(-1)+1=-4+3+1=0$ | M1 | 2.1 | **DR**<br>Use of f(-1) must be seen. Do not allow for algebraic divison. | Allow without conclusion if preceded by "If $f(-1)=0$ then $(x+1)$ will be a factor" or similar |
| | | | Therefore $(x+1)$ is a factor | A1<br>[2] | 2.2a | Clear conclusion must be made | |
| | ii | | $f(x)=(x+1)(4x^2-4x+1)=0$ | M1 | 1.1a | **DR**<br>Attempt to divide or to factorise by inspection with $4x^2$ | Allow full credit for $(x+1)(4x-2)(x-0.5)$ |
| | | | | A1 | 1.1b | correct quadratic factor seen or implied by correct linear factors | |
| | | | $=(x+1)(2x-1)^2=0$ | | | | No marks for solving the cubic on the calculator |
| | | | $x=-1,\ \frac{1}{2}$ [repeated] | A1<br>[3] | 1.1b | Both roots seen derived from 3 correct linear factors or use of quadratic formula | |